CAA03 | Study Guide

Fundamentals of Information Systems • Balanced revision notes • English B2–C1

Based on the chapter sequence of Rainer & Prince, Introduction to Information Systems, 9th ed., International Adaptation (Wiley, 2022). Original explanations; not a reproduction of the textbook.

Contents

- Chapter 11. Customer Relationship Management and Supply Chain Management
- Chapter 12. Business Analytics
- Chapter 13. Acquiring Information Systems and Applications
- Chapter 14. Artificial Intelligence

How to study

Read each chapter overview, explain every highlighted definition aloud, follow the concept map, then answer the questions without looking at the key. Compare your answer with the model answer and revisit weak areas.

Scope note: this guide follows the verified chapter titles and general concepts, not the complete text or UOC-specific grading rubric. Use assigned cases and course instructions for assessment preparation.

Chapter 11 | Customer Relationship Management and Supply Chain Management

COORDINATING DEMAND AND RELATIONSHIPS

Learning outcomes

After studying this chapter, you should be able to define the core vocabulary, explain how the concepts operate in an organisation, apply them to a practical situation and evaluate a trade-off.

Conceptual overview

Customer Relationship Management and Supply Chain Management concerns decisions that connect organisational goals with information, processes and technology. The central question is not simply which tools exist, but how they create value, what assumptions they require and what risks their use introduces. The concepts below provide a practical vocabulary for analysing realistic organisational cases.

Essential definitions

- **CRM** — Strategies and systems for acquiring, serving and retaining customers.
- **SCM** — Coordination of material, information and financial flows across supply networks.
- **Customer lifetime value** — Estimate of the value a customer relationship generates over its duration.
- **Bullwhip effect** — Increasing variability in orders upstream from end-customer demand.
- **Supply chain visibility** — Ability to observe relevant inventory, orders and shipments across participants.

Key ideas and reasoning

1. CRM activities include marketing campaigns, sales pipelines, customer service and retention analysis.
2. SCM covers planning, sourcing, production, distribution and returns; collaboration improves responsiveness.
3. Forecast errors, delays and batch ordering can amplify demand swings across a supply chain.
4. Integration across CRM, ERP and SCM helps connect customer demand with procurement and delivery capacity.

Chapter 11 | Application and analysis

Putting the concepts together

What it means in practice: CRM activities include marketing campaigns, sales pipelines, customer service and retention analysis. To assess this point, identify the actors involved, the information they require, the process being changed and a measurable result. Avoid assuming that installing software automatically produces a benefit.

Why it matters: SCM covers planning, sourcing, production, distribution and returns; collaboration improves responsiveness. To assess this point, identify the actors involved, the information they require, the process being changed and a measurable result. Avoid assuming that installing software automatically produces a benefit.

A management implication: Forecast errors, delays and batch ordering can amplify demand swings across a supply chain. To assess this point, identify the actors involved, the information they require, the process being changed and a measurable result. Avoid assuming that installing software automatically produces a benefit.

A critical perspective: Integration across CRM, ERP and SCM helps connect customer demand with procurement and delivery capacity. To assess this point, identify the actors involved, the information they require, the process being changed and a measurable result. Avoid assuming that installing software automatically produces a benefit.

Worked business example

Electronics distributor. Customer service sees delivery delays from logistics; procurement adjusts replenishment using demand and stock data.

Analysis: identify the input data, the responsible users, the system actions and the outcome. Then consider one failure mode, such as inaccurate records, poor adoption, excessive cost or insufficient security. Propose a metric to test whether the solution actually improves performance.

Concept map

Demand-to-delivery chain
Customer demand → CRM / Sales → Planning → Suppliers → Warehouse → Delivery

Connections and common mistakes

Connections: link this chapter to strategy (Chapter 2), ethics (Chapter 3), security (Chapter 4) and data quality (Chapter 5). Even a narrow operational tool has implications for these wider areas.

Common mistake: memorising a term without explaining its purpose. In an assessment, define the term, describe how it works, apply it to the case and justify the likely benefit or risk.

Chapter 11 | Revision and answer key

Short-answer questions

1. How do CRM and SCM differ?

2. What causes the bullwhip effect?

3. Why does customer data matter to supply planning?

Model answers

1. CRM focuses on customer relationships; SCM coordinates supply and fulfilment.

2. Distorted demand signals, lead times and ordering policies amplify variation.

3. It improves forecasts and aligns supply with expected demand.

Mini case: make a recommendation

An organisation plans to invest in a digital solution related to customer relationship management and supply chain management. Its managers want faster processes, fewer errors and improved service, but its employees worry about training, privacy and additional workload.

Task: recommend two actions before implementation and two indicators for evaluation. Explain one trade-off. Suggested approach: map the current process and stakeholder needs; identify data and security requirements; choose indicators such as cycle time, error rate, satisfaction or total cost; and review outcomes after a pilot.

Self-check checklist

☐ I can define each highlighted term in my own words.

☐ I can explain the concept map without reading it.

☐ I can apply at least two concepts to a new business case.

☐ I can identify one benefit, one limitation and one measurable KPI.

☐ I have answered the revision questions before checking the key.

Further reading

Consult the corresponding chapter in the UOC-assigned Wiley textbook, including its opening/closing cases and glossary. Student companion resources: https://bcs.wiley.com/he-bcs/Books?action=index&bcsId=12246&itemId=111985993X

Chapter 12 | Business Analytics

USING EVIDENCE TO IMPROVE DECISIONS

Learning outcomes

After studying this chapter, you should be able to define the core vocabulary, explain how the concepts operate in an organisation, apply them to a practical situation and evaluate a trade-off.

Conceptual overview

Business Analytics concerns decisions that connect organisational goals with information, processes and technology. The central question is not simply which tools exist, but how they create value, what assumptions they require and what risks their use introduces. The concepts below provide a practical vocabulary for analysing realistic organisational cases.

Essential definitions

- **Business intelligence** — Methods and systems that turn data into reporting and decision-relevant information.
- **Descriptive analytics** — Analysis of what has happened.
- **Predictive analytics** — Methods that estimate likely future outcomes.
- **Prescriptive analytics** — Methods that suggest actions given objectives and constraints.
- **KPI** — Quantified indicator used to assess progress toward a defined goal.

Key ideas and reasoning

5. An analytics workflow frames the business question, acquires and cleans data, analyses patterns, communicates findings and evaluates decisions.
6. Dashboards should show relevant metrics, comparisons and context rather than every available measure.
7. Correlation alone does not establish causation; sampling bias and missing data can mislead decision-makers.
8. Model performance and business impact are different: a statistically strong forecast may still be unsuitable for an operational decision.

Chapter 12 | Application and analysis

Putting the concepts together

What it means in practice: An analytics workflow frames the business question, acquires and cleans data, analyses patterns, communicates findings and evaluates decisions. To assess this point, identify the actors involved, the information they require, the process being changed and a measurable result. Avoid assuming that installing software automatically produces a benefit.

Why it matters: Dashboards should show relevant metrics, comparisons and context rather than every available measure. To assess this point, identify the actors involved, the information they require, the process being changed and a measurable result. Avoid assuming that installing software automatically produces a benefit.

A management implication: Correlation alone does not establish causation; sampling bias and missing data can mislead decision-makers. To assess this point, identify the actors involved, the information they require, the process being changed and a measurable result. Avoid assuming that installing software automatically produces a benefit.

A critical perspective: Model performance and business impact are different: a statistically strong forecast may still be unsuitable for an operational decision. To assess this point, identify the actors involved, the information they require, the process being changed and a measurable result. Avoid assuming that installing software automatically produces a benefit.

Worked business example

Subscription service. A dashboard describes churn, a model estimates cancellation risk, and a campaign optimiser selects cost-effective retention offers.

Analysis: identify the input data, the responsible users, the system actions and the outcome. Then consider one failure mode, such as inaccurate records, poor adoption, excessive cost or insufficient security. Propose a metric to test whether the solution actually improves performance.

Concept map

Analytics maturity
Descriptive: what? → Diagnostic: why? → Predictive: what next? → Prescriptive: what to do?

Connections and common mistakes

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Common mistake: memorising a term without explaining its purpose. In an assessment, define the term, describe how it works, apply it to the case and justify the likely benefit or risk.

Chapter 12 | Revision and answer key

Short-answer questions

1. Distinguish predictive from prescriptive analytics.

2. Why is data cleaning important?

3. What is a limitation of correlation?

Model answers

1. Predictive estimates outcomes; prescriptive recommends actions.

2. Errors and inconsistencies can distort findings and models.

3. Two variables may move together without one causing the other.

Mini case: make a recommendation

An organisation plans to invest in a digital solution related to business analytics. Its managers want faster processes, fewer errors and improved service, but its employees worry about training, privacy and additional workload.

Task: recommend two actions before implementation and two indicators for evaluation. Explain one trade-off. Suggested approach: map the current process and stakeholder needs; identify data and security requirements; choose indicators such as cycle time, error rate, satisfaction or total cost; and review outcomes after a pilot.

Self-check checklist

☐ I can define each highlighted term in my own words.

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Chapter 13 | Acquiring Information Systems and Applications

CHOOSING AND DELIVERING EFFECTIVE SYSTEMS

Learning outcomes

After studying this chapter, you should be able to define the core vocabulary, explain how the concepts operate in an organisation, apply them to a practical situation and evaluate a trade-off.

Conceptual overview

Acquiring Information Systems and Applications concerns decisions that connect organisational goals with information, processes and technology. The central question is not simply which tools exist, but how they create value, what assumptions they require and what risks their use introduces. The concepts below provide a practical vocabulary for analysing realistic organisational cases.

Essential definitions

- **SDLC** — Structured stages used to plan, analyse, design, implement and maintain systems.
- **Requirements** — Statements of capabilities, qualities and constraints a solution must satisfy.
- **Feasibility analysis** — Assessment of technical, economic, operational and scheduling viability.
- **Outsourcing** — Using an external provider to deliver specified services or systems.
- **Agile development** — Iterative and collaborative delivery that incorporates feedback and adaptation.

Key ideas and reasoning

9. Build, buy, subscribe and outsource options differ in cost, flexibility, risk, control and speed.
10. Typical lifecycle activities include initiation, requirements, design, development or configuration, testing, deployment and maintenance.
11. Stakeholder involvement, change management and user training are as important as technical implementation.
12. Projects must manage scope, budget, schedule, quality, cybersecurity, vendor dependency and organisational adoption.

Chapter 13 | Application and analysis

Putting the concepts together

What it means in practice: Build, buy, subscribe and outsource options differ in cost, flexibility, risk, control and speed. To assess this point, identify the actors involved, the information they require, the process being changed and a measurable result. Avoid assuming that installing software automatically produces a benefit.

Why it matters: Typical lifecycle activities include initiation, requirements, design, development or configuration, testing, deployment and maintenance. To assess this point, identify the actors involved, the information they require, the process being changed and a measurable result. Avoid assuming that installing software automatically produces a benefit.

A management implication: Stakeholder involvement, change management and user training are as important as technical implementation. To assess this point, identify the actors involved, the information they require, the process being changed and a measurable result. Avoid assuming that installing software automatically produces a benefit.

A critical perspective: Projects must manage scope, budget, schedule, quality, cybersecurity, vendor dependency and organisational adoption. To assess this point, identify the actors involved, the information they require, the process being changed and a measurable result. Avoid assuming that installing software automatically produces a benefit.

Worked business example

University appointment application. The team compares a SaaS booking platform with a custom system, defines accessibility requirements and pilots a solution.

Analysis: identify the input data, the responsible users, the system actions and the outcome. Then consider one failure mode, such as inaccurate records, poor adoption, excessive cost or insufficient security. Propose a metric to test whether the solution actually improves performance.

Concept map

System delivery lifecycle
Need → Feasibility → Requirements → Design → Build/Buy → Test → Deploy → Improve

Connections and common mistakes

Connections: link this chapter to strategy (Chapter 2), ethics (Chapter 3), security (Chapter 4) and data quality (Chapter 5). Even a narrow operational tool has implications for these wider areas.

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Chapter 13 | Revision and answer key

Short-answer questions

1. What distinguishes functional and non-functional requirements?

2. When might buying software be preferable to building it?

3. Why is user acceptance testing useful?

Model answers

1. Functional requirements specify behaviours; non-functional requirements specify qualities such as performance and accessibility.

2. When standard functionality meets needs at lower total cost and risk.

3. It checks whether real workflows and user expectations are satisfied.

Mini case: make a recommendation

An organisation plans to invest in a digital solution related to acquiring information systems and applications. Its managers want faster processes, fewer errors and improved service, but its employees worry about training, privacy and additional workload.

Task: recommend two actions before implementation and two indicators for evaluation. Explain one trade-off. Suggested approach: map the current process and stakeholder needs; identify data and security requirements; choose indicators such as cycle time, error rate, satisfaction or total cost; and review outcomes after a pilot.

Self-check checklist

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Chapter 14 | Artificial Intelligence

CAPABILITIES, APPLICATIONS AND RESPONSIBLE ADOPTION

Learning outcomes

After studying this chapter, you should be able to define the core vocabulary, explain how the concepts operate in an organisation, apply them to a practical situation and evaluate a trade-off.

Conceptual overview

Artificial Intelligence concerns decisions that connect organisational goals with information, processes and technology. The central question is not simply which tools exist, but how they create value, what assumptions they require and what risks their use introduces. The concepts below provide a practical vocabulary for analysing realistic organisational cases.

Essential definitions

- **Artificial intelligence** — Field concerned with systems performing tasks associated with perception, reasoning, learning or language.
- **Machine learning** — Approaches that learn patterns from examples rather than relying only on explicitly coded rules.
- **Deep learning** — Machine learning using multilayer neural networks.
- **Natural language processing** — Computational methods for analysing, understanding or generating human language.
- **Expert system** — Knowledge-based software using encoded rules and facts to support specialised decisions.

Key ideas and reasoning

13. Supervised learning uses labelled examples; unsupervised learning discovers structure in unlabelled data; reinforcement learning optimises actions through feedback.
14. Business uses include recommendations, forecasting, document classification, anomaly detection and customer support.
15. AI deployment requires suitable data, evaluation against business goals, human oversight and monitoring for drift.
16. Risks include biased decisions, lack of explainability, privacy problems, unreliable outputs and excessive automation.

Chapter 14 | Application and analysis

Putting the concepts together

What it means in practice: Supervised learning uses labelled examples; unsupervised learning discovers structure in unlabelled data; reinforcement learning optimises actions through feedback. To assess this point, identify the actors involved, the information they require, the process being changed and a measurable result. Avoid assuming that installing software automatically produces a benefit.

Why it matters: Business uses include recommendations, forecasting, document classification, anomaly detection and customer support. To assess this point, identify the actors involved, the information they require, the process being changed and a measurable result. Avoid assuming that installing software automatically produces a benefit.

A management implication: AI deployment requires suitable data, evaluation against business goals, human oversight and monitoring for drift. To assess this point, identify the actors involved, the information they require, the process being changed and a measurable result. Avoid assuming that installing software automatically produces a benefit.

A critical perspective: Risks include biased decisions, lack of explainability, privacy problems, unreliable outputs and excessive automation. To assess this point, identify the actors involved, the information they require, the process being changed and a measurable result. Avoid assuming that installing software automatically produces a benefit.

Worked business example

Customer support assistant. A model classifies incoming requests and proposes answers; staff review sensitive cases and monitor error rates.

Analysis: identify the input data, the responsible users, the system actions and the outcome. Then consider one failure mode, such as inaccurate records, poor adoption, excessive cost or insufficient security. Propose a metric to test whether the solution actually improves performance.

Concept map

AI application pipeline
Business problem → Data → Model → Evaluation → Human review → Monitoring

Connections and common mistakes

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Chapter 14 | Revision and answer key

Short-answer questions

1. How are AI, ML and deep learning related?

2. Why is human oversight valuable in high-impact AI?

3. Give one example each of supervised and unsupervised learning.

Model answers

1. ML is a subset of AI; deep learning is a subset of ML.

2. It helps identify errors, bias and harmful decisions before or after deployment.

3. Supervised: spam classification; unsupervised: grouping customers by behaviour.

Mini case: make a recommendation

An organisation plans to invest in a digital solution related to artificial intelligence. Its managers want faster processes, fewer errors and improved service, but its employees worry about training, privacy and additional workload.

Task: recommend two actions before implementation and two indicators for evaluation. Explain one trade-off. Suggested approach: map the current process and stakeholder needs; identify data and security requirements; choose indicators such as cycle time, error rate, satisfaction or total cost; and review outcomes after a pilot.

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CAA03 | Final integrated revision

Compare the four chapters: explain how information systems enable business outcomes, what information they need, how people use them and which ethical or security concerns arise. Draw one end-to-end organisational process and label the systems, stakeholders and performance indicators.

Source and edition

Rainer, R. Kelly, and Brad Prince. Introduction to Information Systems. 9th edition, International Adaptation. Wiley, 2022. Interactive e-text ISBN 9781119860020. Verified contents: https://www.wileydirect.com.au/blog/buy/introduction-to-information-systems-9th-edition-international-adaptation/